2.4 Analysis in $\mathbb{R}^n$: Solutions

Warm-up

1.

For $0 \le x < 1$, $x^n \to 0$; at $x = 1$, $x^n = 1$. So the pointwise limit is $f = 0$ on $[0, 1)$, $f(1) = 1$. The convergence is not uniform on $[0, 1]$: a uniform limit of continuous functions is continuous, and f is not. (Directly: $\sup_{[0,1]} |x^n - f(x)| = \sup_{[0,1)} x^n = 1$.) On $[0, a]$ with $a < 1$, $\sup |x^n - 0| = a^n \to 0$, so the convergence is uniform.

2.

$f_n(0) = 0$. For $x > 0$, $nxe^{-nx^2} = x \cdot \frac{n}{e^{nx^2}} \to 0$ since e^{nx^2} grows faster than n. So $f_n \to 0$ pointwise. But

$$\int_0^1 nxe^{-nx^2} dx = \Big[-\tfrac{1}{2} e^{-nx^2} \Big]_0^1 = \tfrac{1}{2}(1 - e^{-n}) \to \tfrac{1}{2} \neq 0 = \int_0^1 0.$$

The convergence is not uniform: if it were, the integrals would converge to 0. (Directly: $f_n(1/\sqrt{n}) = \sqrt{n}\, e^{-1} \to \infty$.)

3.

$\left|\frac{\sin nx}{n^2}\right| \le \frac{1}{n^2}$ and $\sum \frac{1}{n^2} < \infty$, so by the M-test the series converges uniformly on $\mathbb{R}$; each term is continuous, so the sum is continuous. The differentiated series $\sum \frac{\cos nx}{n}$ at $x = 0$ is $\sum \frac{1}{n}$, which diverges. So the hypothesis "the derivatives converge uniformly" fails, and the term-by-term differentiation theorem cannot be used, at least on any interval containing 0. Uniform convergence of a series does not by itself permit term-by-term differentiation.

4.

(a)
$$Df(x, y) = \begin{pmatrix} 2xy & x^2 \\ y\cos(xy) & x\cos(xy) \\ e^{x+y} & e^{x+y} \end{pmatrix}.$$

(All partials are continuous, so f is differentiable and this is its derivative.)

(b) By the Leibniz formula, $\det(I + H) = \sum_\sigma \operatorname{sgn}\sigma \prod_i (\delta_{i\sigma(i)} + h_{i\sigma(i)})$. The identity permutation contributes $\prod_i (1 + h_{ii}) = 1 + \sum_i h_{ii} +$ (products of at least two entries of H). A permutation $\sigma \neq \mathrm{id}$ moves at least two indices i, and for those $\delta_{i\sigma(i)} = 0$, so its term is a product containing at least two entries of H. Hence

$$\det(I + H) = 1 + \operatorname{tr} H + R(H),$$

where R is a polynomial in the entries of H with every monomial of degree at least 2. Since each entry satisfies $|h_{ij}| \le \|H\|$, for $\|H\| \le 1$ we get $|R(H)| \le C\|H\|^2$ for a constant C (depending only on n). So $\det(I + H) - \det(I) - \operatorname{tr} H = o(\|H\|)$, i.e. $D(\det)(I)H = \operatorname{tr} H$.

5.

$f(x, 0) = 0$ for all x, so $\partial_x f(0, 0) = 0$; similarly $\partial_y f(0, 0) = 0$. But $f(t, t) = \frac{1}{2}$ for $t \neq 0$, so $f(t, t) \to \frac{1}{2} \neq f(0, 0)$: f is not continuous at 0, hence not differentiable there.

6.

Fix $a \in U$ and let $S = \{x \in U : f(x) = f(a)\}$. S is non-empty and closed in U (as f is continuous). S is open: if $x \in S$, choose a ball $B \subseteq U$ around x; B is convex and $\|Df\| = 0$ on B, so the mean value inequality gives $|f(z) - f(x)| \leq 0 \cdot |z - x| = 0$ for $z \in B$, i.e. $B \subseteq S$. Since U is connected, $S = U$.

Core

7.

For $f \in \mathcal{F}$ and $x, y \in [0,1]$, the mean value theorem gives $|f(x) - f(y)| \leq |x - y|$. So $\mathcal{F}$ is equicontinuous (take $\delta = \varepsilon$), and $|f(x)| = |f(x) - f(0)| \leq x \leq 1$, so $\mathcal{F}$ is uniformly bounded. By Arzelà–Ascoli, $\overline{\mathcal{F}}$ is compact.

$\mathcal{F}$ is not closed. Let $g(x) = |x - \frac{1}{2}| - \frac{1}{2}$ and

$$g_n(x) = \sqrt{(x - \tfrac{1}{2})^2 + n^{-2}} - \sqrt{\tfrac{1}{4} + n^{-2}}.$$

Then g_n is C^1, $g_n(0) = 0$, and $|g_n'(x)| = \frac{|x-1/2|}{\sqrt{(x-1/2)^2+n^{-2}}} \leq 1$, so $g_n \in \mathcal{F}$. Since $0 \leq \sqrt{a^2 + \varepsilon^2} - |a| \leq \varepsilon$, we get $|g_n(x) - g(x)| \leq \frac{2}{n}$, so $g_n \to g$ uniformly. But g is not differentiable at $\frac{1}{2}$, so $g \notin \mathcal{F}$.

8.

$|\sin nx| \leq 1$, so the sequence is bounded. For $m \neq n$,

$$\int_0^\pi (\sin mx - \sin nx)^2 dx = \int_0^\pi \sin^2 mx\, dx + \int_0^\pi \sin^2 nx\, dx - 2\int_0^\pi \sin mx \sin nx\, dx = \frac{\pi}{2} + \frac{\pi}{2} - 0 = \pi,$$

using $2 \sin mx \sin nx = \cos(m-n)x - \cos(m+n)x$, which integrates to 0 over $[0, \pi]$ for $m \neq n$ (both $m \pm n$ are non-zero integers). If a subsequence f_{n_k} converged uniformly, it would be uniformly Cauchy, and

$$\pi = \int_0^\pi (f_{n_k} - f_{n_l})^2 \leq \pi \|f_{n_k} - f_{n_l}\|_\infty^2 \to 0,$$

a contradiction. Arzelà–Ascoli then says the family cannot be equicontinuous, and indeed it is not: $f_n(0) = 0$ but $f_n\left(\frac{\pi}{2n}\right) = 1$, with $\frac{\pi}{2n} \to 0$.

9.

By linearity, $\int_0^1 fp = 0$ for every polynomial p. By the Weierstrass approximation theorem choose polynomials $p_n \to f$ uniformly on $[0,1]$. Then

$$\int_0^1 f^2 = \int_0^1 f(f - p_n) + \int_0^1 f p_n = \int_0^1 f(f - p_n),$$

and $\left|\int_0^1 f(f - p_n)\right| \leq \|f\|_\infty \|f - p_n\|_\infty \to 0$. So $\int_0^1 f^2 = 0$. Since f^2 is continuous and non-negative, $f \equiv 0$.

10.

$\mathcal{A} = \{p(x^2)\}$ is a subalgebra of $C[0,1]$ containing the constants. It separates points of $[0,1]$: if $x \neq y$ in $[0,1]$ then $x^2 \neq y^2$, so the function $x^2 \in \mathcal{A}$ separates them. By Stone–Weierstrass, $\mathcal{A}$ is dense in $C[0,1]$.

On $[-1,1]$ every element of $\mathcal{A}$ is even. If $h_n \to h$ uniformly (even pointwise) with all h_n even, then $h(-x) = \lim h_n(-x) = \lim h_n(x) = h(x)$, so h is even. The function x is not even, so it is not in the closure of $\mathcal{A}$. (Stone–Weierstrass fails because $\mathcal{A}$ does not separate x from $-x$.)

11.

Let $T(x,y) = \big(\frac{1}{4}\sin(x+y),\ \frac{1}{4}\cos(x-y)\big)$ on $\mathbb{R}^2$ with the norm $\|(x,y)\|_\infty = \max(|x|,|y|)$, which makes $\mathbb{R}^2$ complete. Since $|\sin a - \sin b| \le |a-b|$ and $|\cos a - \cos b| \le |a-b|$,

$$|T_1(x,y) - T_1(x',y')| \le \tfrac{1}{4}(|x-x'| + |y-y'|) \le \tfrac{1}{2}\|(x,y) - (x',y')\|_\infty,$$

and the same bound holds for T_2. So T is a contraction with constant $\frac{1}{2}$, and by the contraction mapping theorem it has exactly one fixed point, which is exactly one solution of the system.

12.

(a) Let $X = C[0,h]$ with the sup norm (complete). A C^1 function y solves the problem iff it is continuous and satisfies

$$y(t) = y_0 + \int_0^t F(s, y(s))\, ds \qquad (0 \le t \le h).$$

(If y solves the ODE, integrate. Conversely, if y is continuous and satisfies the integral equation, the integrand $s \mapsto F(s,y(s))$ is continuous, so by the fundamental theorem of calculus y is C^1 with $y' = F(t,y)$, and $y(0) = y_0$.) Define $T : X \to X$ by $(Ty)(t) = y_0 + \int_0^t F(s,y(s))\, ds$; Ty is continuous (indeed C^1). For $y, z \in X$ and $t \in [0,h]$,

$$|(Ty)(t) - (Tz)(t)| \le \int_0^t |F(s,y(s)) - F(s,z(s))|\, ds \le tL\|y-z\|_\infty \le hL\|y-z\|_\infty.$$

Since $hL < 1$, T is a contraction and has exactly one fixed point: exactly one solution on $[0,h]$.

(b) $(Ty)(t) = 1 + \int_0^t y(s)\, ds$. Starting from $y_0 = 1$: $y_1 = 1+t$, $y_2 = 1 + t + \frac{t^2}{2}$, and by induction $y_n(t) = \sum_{k=0}^n \frac{t^k}{k!}$. These converge uniformly on bounded intervals to e^t, the solution.

13.

(a) $(A+H)^2 - A^2 - (AH+HA) = H^2$ and $\|H^2\| \le \|H\|^2 = o(\|H\|)$. So $Df(A)H = AH + HA$.

(b) *Openness.* Let A be invertible and $\|H\| < \frac{1}{\|A^{-1}\|}$. Then $A + H = A(I + A^{-1}H)$ with $\|A^{-1}H\| < 1$, and $I + E$ with $\|E\| < 1$ is invertible, with inverse given by the Neumann series $\sum_{k\ge 0}(-E)^k$ and $\|(I+E)^{-1}\| \le \frac{1}{1-\|E\|}$. So $A + H$ is invertible: $GL_n(\mathbb{R})$ is open. Moreover, if $\|H\| \le \frac{1}{2\|A^{-1}\|}$, then $\|(A+H)^{-1}\| \le \|(I + A^{-1}H)^{-1}\|\,\|A^{-1}\| \le 2\|A^{-1}\|$.

Derivative. From $(A+H)^{-1} - A^{-1} = -(A+H)^{-1}HA^{-1}$ (multiply out: $(A+H)^{-1}[A - (A+H)]A^{-1}$),

$$(A+H)^{-1} - A^{-1} + A^{-1}HA^{-1} = \big(A^{-1} - (A+H)^{-1}\big)HA^{-1} = (A+H)^{-1}HA^{-1}HA^{-1}.$$

Its norm is at most $2\|A^{-1}\|^3\|H\|^2 = o(\|H\|)$. So $Dg(A)H = -A^{-1}HA^{-1}$.

14.

f is C^∞ with

$$Df(x,y) = \begin{pmatrix} 2x & -2y \\ 2y & 2x \end{pmatrix}, \qquad \det Df = 4(x^2+y^2).$$

For $p \neq 0$ this is non-zero, so by the inverse function theorem f restricts to a C^1 (indeed C^∞) diffeomorphism from a neighbourhood of p onto an open set. But $f(-x,-y) = f(x,y)$, so f is not injective on $\mathbb{R}^2 \setminus \{0\}$. (In complex notation $f(z) = z^2$.) At 0, $Df(0) = 0$, and f is not injective on any neighbourhood of 0 (again $f(p) = f(-p)$), so there is no local inverse there.

15.

Let $F(x,y,z) = x^2 + 2y^2 + z^3 + xz - 5$, a C^∞ function with $F(1,1,1) = 1+2+1+1-5 = 0$. Its partials are $F_x = 2x + z$, $F_y = 4y$, $F_z = 3z^2 + x$; at $(1,1,1)$ these are 3, 4, 4. Since $F_z(1,1,1) = 4 \neq 0$, the implicit function theorem gives a C^1 function g near $(1,1)$ with $g(1,1) = 1$ and $F(x,y,g(x,y)) = 0$, describing all solutions near $(1,1,1)$. Differentiating,

$$\frac{\partial g}{\partial x}(1,1) = -\frac{F_x}{F_z} = -\frac{3}{4}, \qquad \frac{\partial g}{\partial y}(1,1) = -\frac{F_y}{F_z} = -1.$$

16.

(a) Since $\nabla g(p) \neq 0$, after relabelling coordinates $\partial_n g(p) \neq 0$. Write $x = (x', x_n)$. By the implicit function theorem there are a neighbourhood V of p' and a C^1 function h on V such that, near p, M is the graph $x_n = h(x')$. Let $v \in \mathbb{R}^n$ with $\nabla g(p) \cdot v = 0$, and define $\gamma(t) = (p' + tv',\ h(p' + tv'))$ for small t. Then $\gamma(t) \in M$, $\gamma(0) = p$, and differentiating $g(x', h(x')) = 0$ gives $\partial_n g(p)\, Dh(p')v' = -\sum_{i<n} \partial_i g(p) v_i = \partial_n g(p) v_n$, so $\gamma'(0) = (v', Dh(p')v') = v$. Since $f \circ \gamma$ has a local maximum at $t = 0$, $0 = (f\circ\gamma)'(0) = \nabla f(p) \cdot v$.

So $\nabla f(p)$ is orthogonal to every vector orthogonal to $\nabla g(p)$, i.e. $\nabla f(p) \in (\nabla g(p)^\perp)^\perp = \operatorname{span} \nabla g(p)$: $\nabla f(p) = \lambda \nabla g(p)$.

(b) The ellipse $M = \{g = 0\}$, $g = \frac{x^2}{4} + y^2 - 1$, is compact, so xy attains a maximum and a minimum on it. $\nabla g = (x/2, 2y)$ vanishes only at $0 \notin M$. By (a) (applied to f and $-f$), at an extremum $(y, x) = \lambda(x/2, 2y)$, i.e. $y = \lambda x/2$ and $x = 2\lambda y$. Then $y = \lambda^2 y$. If $y = 0$ then $x = 0$, not on M; so $\lambda^2 = 1$.

- $\lambda = 1$: $x = 2y$, and $\frac{4y^2}{4} + y^2 = 1$ gives $y = \pm\frac{1}{\sqrt{2}}$, $xy = 2y^2 = 1$.
- $\lambda = -1$: $x = -2y$, $y = \pm\frac{1}{\sqrt{2}}$, $xy = -1$.

So the maximum is 1 (at $\pm(\sqrt{2}, \frac{1}{\sqrt{2}})$) and the minimum is -1 (at $\pm(\sqrt{2}, -\frac{1}{\sqrt{2}})$).

17.

For fixed $x > 0$, $\frac{\partial}{\partial y}\frac{y}{x^2+y^2} = \frac{x^2+y^2-2y^2}{(x^2+y^2)^2} = f(x,y)$, so

$$\int_0^1 f(x,y)\,dy = \frac{1}{x^2+1}, \qquad \int_0^1 \frac{dx}{1+x^2} = \frac{\pi}{4}.$$

Since $f(y,x) = -f(x,y)$, the other iterated integral is $-\frac{\pi}{4}$.

Fubini requires f to be integrable on the square. f is unbounded near 0, so it is not Riemann integrable on $[0,1]^2$; even as an improper (or Lebesgue) integral it fails, because $\iint |f| = \infty$. (In polar coordinates $|f| = \frac{|\cos 2\theta|}{r^2}$, and $\int_0^{\cdot} \frac{1}{r^2} r\,dr$ diverges at 0.) So there is no contradiction.

18.

For $R > 0$ let $I(R) = \int_{-R}^{R} e^{-x^2} dx$, which increases to the convergent improper integral $I = \int_{-\infty}^{\infty} e^{-x^2} dx$ (convergent since $e^{-x^2} \le e^{-|x|}$ for $|x| \ge 1$).

Fubini. $e^{-(x^2+y^2)}$ is continuous on the square $S_R = [-R,R]^2$, so

$$\iint_{S_R} e^{-(x^2+y^2)}\,dA = \int_{-R}^{R}\int_{-R}^{R} e^{-x^2}e^{-y^2}\,dy\,dx = I(R)^2.$$

Sandwich. With D_ρ the closed disc of radius ρ, $D_R \subseteq S_R \subseteq D_{R\sqrt{2}}$, and the integrand is positive, so

$$\iint_{D_R} e^{-|z|^2} \le I(R)^2 \le \iint_{D_{R\sqrt{2}}} e^{-|z|^2}.$$

Change of variables. Polar coordinates $g(r,\theta) = (r\cos\theta, r\sin\theta)$ give a C^1 diffeomorphism from $(0,\rho)\times(0,2\pi)$ onto D_ρ minus the segment $\{(x,0) : 0 \le x \le \rho\}$ and the boundary circle, which have content zero and do not affect the integral; $|\det Dg| = r$. So

$$\iint_{D_\rho} e^{-|z|^2} dA = \int_0^{2\pi}\int_0^{\rho} e^{-r^2} r\,dr\,d\theta = \pi(1 - e^{-\rho^2}).$$

Hence $\pi(1-e^{-R^2}) \le I(R)^2 \le \pi(1-e^{-2R^2})$. Letting $R \to \infty$, $I^2 = \pi$, and $I = \sqrt{\pi}$ since $I > 0$.

19.

The map $(x,y) \mapsto (u,v) = (x+y, x-y)$ is linear with inverse $x = \frac{u+v}{2}$, $y = \frac{u-v}{2}$, whose Jacobian determinant is $\det \begin{pmatrix} 1/2 & 1/2 \\ 1/2 & -1/2 \end{pmatrix} = -\frac{1}{2}$, so $|\det| = \frac{1}{2}$. Since $|x| + |y| = \max(|x+y|, |x-y|)$ (check the four sign cases), D corresponds to the square $\{|u| \le 1, |v| \le 1\}$. Therefore

$$\iint_D (x+y)^2 e^{x-y}\,dA = \frac{1}{2}\int_{-1}^{1}\int_{-1}^{1} u^2 e^v\,dv\,du = \frac{1}{2}\cdot\frac{2}{3}\cdot(e - e^{-1}) = \frac{e-e^{-1}}{3}.$$

Challenge

20.

Write $r^2 = x^2 + y^2$ and $\omega = P\,dx + Q\,dy$ with $P = -y/r^2$, $Q = x/r^2$.

(a) $d\omega = dP \wedge dx + dQ \wedge dy = (\partial_x Q - \partial_y P)\,dx \wedge dy$. Now

$$\partial_x Q = \frac{r^2 - 2x^2}{r^4} = \frac{y^2 - x^2}{r^4}, \qquad \partial_y P = -\frac{r^2 - 2y^2}{r^4} = \frac{y^2 - x^2}{r^4}.$$

So $d\omega = 0$.

(b) $c^*\omega = \frac{-\sin t\cdot(-\sin t)+\cos t\cdot\cos t}{1}dt = dt$, so $\int_c \omega = \int_0^{2\pi} dt = 2\pi$. If $\omega = df$ on U for a smooth function f, then $\int_c df = \int_0^{2\pi} (f\circ c)'(t)\,dt = f(c(2\pi)) - f(c(0)) = 0$, a contradiction. So ω is closed but not exact on U.

(c) On U', every point can be written uniquely as $(r\cos\theta, r\sin\theta)$ with $r > 0$, $\theta \in (-\pi, \pi)$. The function $\theta : U' \to (-\pi,\pi)$ is smooth: on $\{x > 0\}$, $\theta = \arctan(y/x)$; on $\{y > 0\}$, $\theta = \frac{\pi}{2} - \arctan(x/y)$; on $\{y < 0\}$, $\theta = -\frac{\pi}{2} - \arctan(x/y)$; these open sets cover U'. On each piece,

$$d\arctan\frac{y}{x} = \frac{1}{1+y^2/x^2}\cdot\frac{x\,dy - y\,dx}{x^2} = \frac{x\,dy - y\,dx}{r^2}, \qquad d\left(-\arctan\frac{x}{y}\right) = -\frac{1}{1+x^2/y^2}\cdot\frac{y\,dx - x\,dy}{y^2} =$$

So $d\theta = \omega$ on U': ω is exact there.

21.

Bijectivity. Fix $y \in \mathbb{R}^n$ and let $\varphi(x) = x - f(x) + y$. Then $D\varphi(x) = I - Df(x)$ has norm $\le \frac{1}{2}$, and $\mathbb{R}^n$ is convex, so by the mean value inequality $|\varphi(x) - \varphi(x')| \le \frac{1}{2}|x - x'|$. Since $\mathbb{R}^n$ is complete, φ has exactly one fixed point, i.e. exactly one x with $f(x) = y$. So f is a bijection.

Lipschitz inverse. Applying the same estimate to $x \mapsto x - f(x)$:

$$|f(x) - f(x')| \ge |x - x'| - |(x - f(x)) - (x' - f(x'))| \ge \tfrac{1}{2}|x - x'|.$$

So $|f^{-1}(y) - f^{-1}(y')| \le 2|y - y'|$.

C^1 inverse. For each x, $\|I - Df(x)\| \le \frac{1}{2} < 1$, so $Df(x) = I - (I - Df(x))$ is invertible (Neumann series). By the inverse function theorem, f has a local C^1 inverse near every point, and these local inverses agree with the global inverse f^{-1} (by injectivity). So f^{-1} is C^1, with $D(f^{-1})(y) = Df(f^{-1}(y))^{-1}$.

22.

K is continuous on the compact square, so it is bounded, $|K| \le M$, and uniformly continuous: given $\varepsilon > 0$ there is $\delta > 0$ with $|K(x,y) - K(x',y)| < \varepsilon$ whenever $|x - x'| < \delta$, for all y.

For $f \in C[0,1]$ and $|x - x'| < \delta$,

$$|Tf(x) - Tf(x')| \le \int_0^1 |K(x,y) - K(x',y)|\,|f(y)|\,dy \le \varepsilon\|f\|_\infty.$$

So Tf is (uniformly) continuous: T maps $C[0,1]$ to $C[0,1]$. For $\|f\|_\infty \le 1$ the same estimate, with the same δ for all such f, shows $\mathcal{F} = \{Tf : \|f\|_\infty \le 1\}$ is equicontinuous; and $|Tf(x)| \le M$ shows it is uniformly bounded. By Arzelà–Ascoli, $\overline{\mathcal{F}}$ is compact. (In the language of module 3.2: T is a compact operator.)

23.

(a) It suffices to prove $\int_{\partial R} P\,dx = -\iint_R \partial_y P\,dA$ and $\int_{\partial R} Q\,dy = \iint_R \partial_x Q\,dA$.

For the first: by Fubini (the integrand is continuous) and the fundamental theorem of calculus in y,

$$-\iint_R \partial_y P\,dA = -\int_a^b \int_c^d \partial_y P(x,y)\,dy\,dx = \int_a^b \big(P(x,c) - P(x,d)\big)\,dx.$$

On ∂R (anticlockwise): along the bottom edge $y = c$, x goes from a to b, contributing $\int_a^b P(x,c)\,dx$; along the top edge $y = d$, x goes from b to a, contributing $-\int_a^b P(x,d)\,dx$; on the vertical edges $dx = 0$. The sum agrees.

For the second: $\iint_R \partial_x Q\,dA = \int_c^d \int_a^b \partial_x Q\,dx\,dy = \int_c^d \left(Q(b,y) - Q(a,y)\right) dy$. On ∂R: the right edge $x = b$ is traversed upwards, contributing $\int_c^d Q(b,y)\,dy$; the left edge $x = a$ downwards, contributing $-\int_c^d Q(a,y)\,dy$; on the horizontal edges $dy = 0$. Again these agree. Adding gives Green's theorem.

(b) Green's theorem needs P, Q to be C^1 on an open set containing all of R. For ω, $P = -y/r^2$ and $Q = x/r^2$ are not defined (indeed unbounded) at 0, which lies in R. So (a) does not apply, and $\int_{\partial R} \omega$ need not be 0. In fact it is 2π. For example, for $R = [-1,1]^2$: on the right side $x = 1$ (traversed upwards), $\omega = \frac{dy}{1+y^2}$, contributing $\arctan 1 - \arctan(-1) = \frac{\pi}{2}$; by the symmetry $(x,y) \mapsto (-y,x)$, which preserves ω, each of the four sides contributes $\frac{\pi}{2}$. If $0 \notin R$, then (a) applies and $\int_{\partial R} \omega = \iint_R 0 = 0$.